

Dirichlet Process-based Robust Clustering using the Median-of-Means Estimator

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Abstract

Clustering stands as one of the most prominent challenges in unsupervised machine learning. Among centroid-based methods, the classic k -means algorithm, based on Lloyd’s heuristic, is widely used. Nonetheless, it is a well-known fact that k -means and its variants face several challenges, including heavy reliance on initial cluster centroids, susceptibility to converging into local minima of the objective function, and sensitivity to outliers and noise in the data. When data contains noise or outliers, the Median-of-Means (MoM) estimator offers a robust alternative for stabilizing centroid-based methods. On a different note, another limitation in many commonly used clustering methods is the need to specify the number of clusters beforehand. Model-based approaches, such as Bayesian nonparametric models, address this issue by incorporating infinite mixture models, which eliminate the requirement for predefined cluster counts. Motivated by these facts, in this article, we propose an efficient and automatic clustering technique by integrating the strengths of model-based and centroid-based methodologies. Our method mitigates the effect of noise on the quality of clustering; while at the same time, estimates the number of clusters. Statistical guarantees on an upper bound of clustering error, and rigorous assessment through simulated and real datasets, suggest the advantages of our proposed method over existing state-of-the-art clustering algorithms.

1 Introduction

Within the fields of machine learning, data mining, and statistics, clustering stands out as a very prominent challenge in the domain of unsupervised learning. Its focus lies in employing methodologies to reveal underlying patterns, called clusters, within datasets. These clusters are defined so that data points grouped within the same cluster demonstrate a degree of internal similarity (Xu and Tian, 2015). Conversely, data points originating from separate clusters are expected to exhibit notable dissimilarity. Typically, data points are portrayed as vectors encompassing variables, referred to as features, in the machine learning community.

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The k -means algorithm (Lloyd, 1982) stands as a classic and extensively used clustering technique. When given a specific number of clusters, say K , the k -means algorithm iterates through two key steps: cluster assignment, where each data point is assigned to the cluster with the nearest centroid based on Euclidean or ℓ_2 distance, and computing the cluster centroids, which involves placing each cluster’s centroid at the sample mean of the points assigned to that cluster over the course of the current iteration. Given a dataset $\mathcal{X} = \{\mathbf{X}_1, \dots, \mathbf{X}_n\}$, the k -means algorithm attempts to partition $\mathcal{X}$ into K mutually exclusive classes by optimizing the objective function:

$$f_{\text{KM}}(\Theta) = \sum_{i=1}^n \min_{1 \leq j \leq K} \|\mathbf{X}_i - \theta_j\|_2^2. \quad (1)$$

where $\Theta = \{\theta_1, \theta_2, \dots, \theta_K\}$ is the set of centroids corresponding to each of the K clusters, and $\|\cdot\|_2$ is the usual ℓ_2 norm. This optimization seeks to minimize the within-cluster variability. Unfortunately, k -means and its variants suffer from several well-documented limitations, such as significant reliance on the initial selection of cluster centroids (Bachem et al., 2017), tendency to converge to suboptimal local minima rather than the global minimum of the objective function (Xu and Lange, 2019), and importantly, high sensitivity to outliers (Zhang et al., 2021). Moreover, k -means performs poorly when the clusters are non-spherical (Ng et al., 2001), and even when the clusters are spherical but with unequal cluster radii and densities (Raykov et al., 2016). Apart from k -means, some popular clustering methods include its improved version k -means++ (Arthur and Vassilvitskii, 2007), as well as k -medians (Bradley et al., 1996; Arora et al., 2000), k -modes (Chaturvedi et al., 2001), k -Harmonic Means (Zhang et al., 1999), etc.

Another major shortcoming of these algorithms used in practice is that most of them explicitly presuppose the number of clusters. The most commonly recognized algorithms such as k -means clustering, spectral clustering (Ng et al., 2001), MinMax k -means clustering (Tzortzis and Likas, 2014), Gaussian mixture models suffer from this issue.

It is well-established in the machine learning community that Bayesian approaches generally offer room for more flexible models in various settings. For instance, the Dirichlet process mixture model (Hjort et al., 2010), which is notably a Bayesian nonparametric model, gives rise to infinite mixture models that do not require the number of clusters in the dataset to be supplied beforehand. (Kulis and Jordan, 2012) considered such an approach that bridges the concepts of k -means and Gaussian mixture models (Bishop and Nasrabadi, 2006; Murphy, 2018). Nevertheless, their method, called DP means, exhibits flexibility in guessing an optimal number of clusters, the algorithm utilizes the cluster average, i.e., arithmetic mean of the data points within the cluster, for centroid updation, compromising its performance specifically on noisy or outlier-laden datasets.

In this article, we address these challenges by fusing two prominent clustering methodologies: centroid-based and model-based. Our proposed algorithm, DP-MoM, is designed to excel in scenarios involving noisy or outlier-affected data, courtesy of its utilization of the Median-of-Means (MoM) estimator (Nemirovsky and Yudin, 1983; Devroye et al., 2016). Additionally, DP-MoM offers the advantage of not necessitating a predefined number of clusters. To demonstrate the efficacy of our approach, we present a compelling example. We introduce randomly generated noisy observations into the *Jain* dataset (Jain and Law, 2005) (refer to the Experiments section for detailed information) and subsequently apply various cutting-edge algorithms, including our proposed DP-MoM, to evaluate

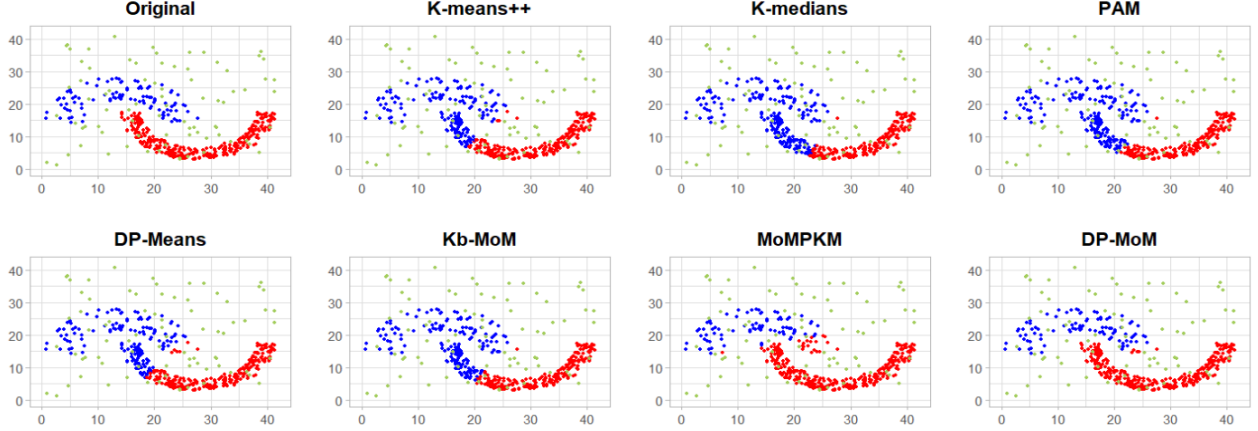


Figure 1: Several state-of-the-art clustering methods fail to achieve proper clustering in presence of noisy observations (light green in color), while the performance of DP-MoM, our proposed algorithm, is nearly optimal.

their respective performances on the original dataset. As illustrated in Figure 1, DP-MoM showcases notably superior clustering accuracy compared to existing algorithms.

2 Background

2.1 Clustering based on Dirichlet Process

Kulis and Jordan (2012) introduced a method utilizing Gibbs sampling, serving as a Bayesian counterpart to representing k -means with a mixture of Gaussians. We assume the following model to capture the cluster structure of the dataset and whose limiting case reduces to the Dirichlet Process Mixture models:

$$\begin{aligned}\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_k &\sim G_0, \quad \pi \sim \text{Dir}(k, \pi_0), \\ z_1, \dots, z_n &\sim \text{Discrete}(\pi), \\ \mathbf{X}_i &\sim \mathcal{N}(\boldsymbol{\theta}_{z_i}, \sigma I) \quad \forall i = 1, 2, \dots, n\end{aligned}$$

where $\boldsymbol{\theta}_j$'s are the cluster centroids, G_0 is taken to be a $\mathcal{N}(0, \rho \mathbf{I})$ prior where I denotes the identity matrix of appropriate order. $\text{Dir}(k, \pi_0)$ denotes the Dirichlet distribution, where π is the mixture probability with $\pi_0 = \frac{\alpha}{k} \mathbf{1}$. Here, $\mathbf{1}$ denotes the vector (of appropriate order) of all 1's. For $i = 1, 2, \dots, n$, z_i denotes the label assigned to the data points $\mathbf{X}_i$, and $\text{Discrete}(\pi)$ indicates that z_i takes the value j with probability π_j , for $j = 1, \dots, k$.

The hard clustering algorithm, called *DP-means*, proposed by Kulis and Jordan (2012) is essentially the case when $\sigma \rightarrow 0$. This limiting case boils down to minimizing the objective:

$$f_{\text{DP}}(\boldsymbol{\Theta}, k) = \sum_{i=1}^n \min_{1 \leq j \leq k} \|\mathbf{X}_i - \boldsymbol{\theta}_j\|_2^2 + \lambda k. \quad (2)$$

The minimization of the function in (2) with respect to k is performed iteratively. At each step of the algorithm, the distance from each data point to its closest cluster centroid is determined.

Subsequently, each point is assigned to the cluster with the closest centroid, unless this distance exceeds λ . In such an instance, a new cluster is initialized with the data point serving as the centroid of the newly created cluster. This algorithm determines the number of clusters in a dataset without necessitating prior knowledge of k . It maintains the simplicity inherent in Lloyd's approach, while ensuring effectiveness even in situations where the true number of clusters is unknown.

2.2 Median-of-Means (MoM)

Let us consider a simple scenario first where we observe $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \sim F$ with a goal to estimate the mean of the distribution F , i.e., $\mu_0 = \mathbb{E}[\mathbf{X}_1] = \int x dF(x)$.

We employ the median-of-means (MoM) estimator to estimate μ_0 as follows: Assume that the sample size $n = bL$ where L is the number of buckets (disjoint subsamples) and b is the size of each bucket. We first split the data randomly into L partitions (or buckets), and calculate the mean of the data points belonging to each partition. This gives rise to estimators $\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_L$. The MoM estimator defined to be the median of these b many mean estimators, namely,

$$\hat{\mu}_{\text{MoM}} = \text{median}(\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_L). \quad (3)$$

As opposed to the sample mean which has a finite sample breakdown point of 0, the MoM estimator is far more robust, exhibiting a breakdown point of $[(L-1)/2]/n$, thereby guaranteeing stability when the number of buckets is of the order of the number of data points i.e. $L = \mathcal{O}(n)$. Another reason why this estimator is of such interest (apart from being a robust estimator) is that given $\text{var}(\mathbf{X}_1) = \sigma^2 < \infty$ in the finite sample case, it satisfies concentration inequalities that aid in proving consistency results thereby leading to stable and consistent MoM-based estimators.

In case of centroid-based clustering, the median-of-means estimator is used as follows: We first partition the dataset into L subparts. In each iteration, we calculate the mean objective function value for each bucket l , $\frac{1}{b} \sum_{i \in B_l} f_{\Theta}(\mathbf{X}_i)$ (where Θ is the collection of centroids from the previous iteration) and choose the bucket L_t in such a way that

$$\frac{1}{b} \sum_{i \in B_{L_t}} f_{\Theta}(\mathbf{X}_i) = \text{median}_{1 \leq j \leq L} \left[\frac{1}{b} \sum_{i \in B_j} f_{\Theta}(\mathbf{X}_i) \right]. \quad (4)$$

The centroids Θ are recalculated based on the observations in the bucket B_{L_t} and all the observations are clustered using these centroids.

3 Dirichlet Process Clustering with MoM

3.1 Problem Formulation

The problem posed to us is that of partitioning a given dataset $\mathcal{X} = \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\} \subset \mathbb{R}^p$ into natural, disjoint clusters such that the variance within each cluster is minimized at the same time maximizing the inter-partition variability.

In the context of centroid-based clustering, the j^{th} cluster is represented by its centroid θ_j . The concept of "closeness" is quantified by utilization of a Bregman divergence (Bregman, 1967) as a

dissimilarity measure. Let us denote the set of all non-negative real numbers by $\mathbb{R}_0^+$. Any function $\phi : \mathbb{R}^p \rightarrow \mathbb{R}$ that is convex and differentiable, gives rise to the Bregman divergence $d_\phi : \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}_0^+$ defined as

$$d_\phi(\mathbf{x}, \mathbf{y}) = \phi(\mathbf{x}) - \phi(\mathbf{y}) - \langle \nabla \phi(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle. \quad (5)$$

The Bregman divergence that we shall be using in our framework is the Euclidean distance, generated by $\phi(\mathbf{u}) = \|\mathbf{u}\|_2^2$; although it can be generalized to any valid Bregman divergence. Prior knowledge of the number of centroids k enables us to accomplish clustering by minimizing the objective function

$$f_\Theta(\mathbf{X}) := \frac{1}{n} \sum_{i=1}^n \Psi(d_\phi(\mathbf{X}_i, \boldsymbol{\theta}_1), d_\phi(\mathbf{X}_i, \boldsymbol{\theta}_2), \dots, d_\phi(\mathbf{X}_i, \boldsymbol{\theta}_k)). \quad (6)$$

Here, $\Psi : \mathbb{R}_0^{+k} \rightarrow \mathbb{R}_0^+$ is the function $\min_{1 \leq j \leq k} d_\phi(\mathbf{X}, \boldsymbol{\theta}_j)$. In our case, we will seek to minimize the objective function

$$h(\Theta) := \text{median} \left(\frac{1}{b} \sum_{i \in B_1} f_\Theta(\mathbf{X}_i), \frac{1}{b} \sum_{i \in B_2} f_\Theta(\mathbf{X}_i), \dots, \frac{1}{b} \sum_{i \in B_L} f_\Theta(\mathbf{X}_i) \right) + \lambda k \quad (7)$$

with respect to both $\{\boldsymbol{\theta}_j\}_{1 \leq j \leq k}$ and k .

3.2 Optimization

Optimizing the above objective is achieved using gradient-based methods. In our case, we employ the *AdaGrad* algorithm (Duchi et al., 2011) for the said purpose. The centroids are updated as follows:

$$\boldsymbol{\theta}_j^{(t+1)} := \boldsymbol{\theta}_j^{(t)} - \frac{\eta}{\sqrt{\varepsilon + \sum_{t'=1}^t \|g_j^{(t')}\|^2}} \cdot g_j^{(t)}, \quad (8)$$

with

$$g_j^{(t)} = \frac{1}{b} \sum_{i \in B_{t_j}} 2(\boldsymbol{\theta}_j^{(t)} - \mathbf{X}_i) \cdot \mathbf{I}_{\{\mathbf{X}_i \in \mathcal{C}_j\}} \quad (9)$$

where $\mathbf{I}_{\{\mathbf{X}_i \in \mathcal{C}_j\}} = 1$ if $\mathbf{X}_i \in \mathcal{C}_j$, and 0 otherwise.

3.3 Algorithm

Algorithm 1 Dirichlet Process Clustering using Median-of-Means (DP-MoM)

Input: Data matrix $\mathcal{X}$, Penalty parameter λ , ε , Learning Rate η , Tolerance δ .

Output: Number of clusters k , Cluster assignments $\mathcal{U}$, Cluster centroids Θ .

Initialization: Randomly divide $\{1, \dots, n\}$ into L buckets of equal size. Set $\boldsymbol{\theta}_1 = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i$, $k = 1$, $b = \frac{n}{L}$, and $\mathcal{U} = \mathbf{1}$.

- 1: **while** $t < t_{\max}$ or $\left| \frac{h(\Theta^{(t+1)})}{h(\Theta^{(t)})} - 1 \right| > \delta$ **do**
- 2: **for** every observation $\mathbf{X}_i$ **do**
- 3: Compute $a_i = \min \{\|\mathbf{X}_i - \boldsymbol{\theta}_j\|^2, j = 1, \dots, k\}$

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4:   if  $a_i > \lambda$  then
5:       Set  $k = k + 1$ ,  $\boldsymbol{\theta}_k = \mathbf{X}_i$ 
6:       Update  $\mathcal{U}$  by  $u_{ij} = \mathbf{I}_{\{j=k\}}$ 
7:   else
8:       Update  $\mathcal{U}$  by  $u_{ij} = \mathbf{I}_{\{j=\text{argmin}_{1 \leq c \leq k} \|\mathbf{X}_i - \boldsymbol{\theta}_c\|^2\}}$ 
9:   end if
10: end for
11: Find  $l_t \in \{1, 2, \dots, L\}$  such that  $\frac{1}{b} \sum_{i \in B_{l_t}} f_{\boldsymbol{\Theta}^{(t)}}(\mathbf{X}_i) = \text{median}_{1 \leq j \leq L} \left[ \frac{1}{b} \sum_{i \in B_j} f_{\boldsymbol{\Theta}}(\mathbf{X}_i) \right]$ 
12: For each  $j \in \{1, 2, \dots, k\}$ :  $g_j^{(t)} = \frac{1}{b} \sum_{i \in B_{l_t}} 2(\boldsymbol{\theta}_j^{(t)} - \mathbf{X}_i)u_{ij}$ 
13: Update  $\boldsymbol{\Theta}$  by  $\boldsymbol{\theta}_j^{(t+1)} := \boldsymbol{\theta}_j^{(t)} - \frac{\eta}{\sqrt{\varepsilon + \sum_{t'=1}^t \|g_j^{(t')}\|^2}} \cdot g_j^{(t)}$ 
14: end while

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Algorithm 1 summarizes the pseudocode for the above procedure. The tuning parameter ε is set to 1. The learning rate η is typically chosen to be the power of 10 which is of the order of the squared maximum pairwise distance in the dataset, or one lower than that i.e. if the maximum squared separation between any two observations in the data is D , then we set $\eta = 10^{\lceil 2 \log_{10} D \rceil / 2}$ or $10^{\lceil 2 \log_{10} D \rceil / 2 - 1}$ depending on which of these values aids efficient clustering using our proposed method, where $\lceil \cdot \rceil$ represents the ceiling function. Our proposed framework enables us to automatically detect the appropriate number of clusters based on the value of the penalty parameter λ which is optimized by grid-searching.

3.4 Parameter Selection

The first step in our proposed algorithm is partitioning the dataset randomly. This is achieved by choosing a permutation of the data points and then placing them in different buckets in the order of the permutation. Though this technique achieves randomness in terms of partitioning the data, arbitrary partitioning may lead to undesirable results, which is why the partitioning (or permutation for that matter) needs to be carefully chosen. It is pretty obvious that the buckets need to be fairly good representatives of the clusters for us to obtain accurate clustering. For this purpose, in each bucket of size b , we choose b data points using a k -means++ type initialization to choose b centers i.e., each data point has a probability proportional to its distance from the nearest centroid of being chosen. Once they are chosen, for the next bucket, we repeat this process leaving out the data points in all the previous buckets. This indexing scheme gives us a measure of control over the clustering and the subsequent grid-searching to determine the optimal value of the penalty parameter λ and the number of buckets L .

We determine $\lambda_{\min}$ and $\lambda_{\max}$, the minimum and maximum pairwise squared distance between the data points, respectively. 11 equally-spaced points $\lambda_{\min} = \lambda_1^1 < \lambda_2^1 < \dots < \lambda_{10}^1 < \lambda_{11}^1 = \lambda_{\max}$ are picked, and the algorithm is run for these values and for all values of number of partitions, L , such that $2 < L < \frac{n}{3}$. We select $\lambda_{i^*}^1$ corresponding to the most accurate clustering and divide its

neighborhood $[\lambda_{i^*-1}^1, \lambda_{i^*-1}^1]$ ($[\lambda_1^1, \lambda_2^1]$ or $[\lambda_{10}^1, \lambda_{11}^1]$) into 20 divisions and re-run the algorithm as we did in the interval $[\lambda_{\min}, \lambda_{\max}]$. We repeat this one more time, so that the feasible range for the penalty parameter λ has been segmented to the order of 10^3 .

We then choose the λ and L values for which the best clustering accuracy is attained. We call them λ_{opt} and L_{opt} . Since our proposed algorithm is a randomized one, we cannot readily conclude that λ_{opt} and L_{opt} are the only values corresponding to which we will obtain high clustering accuracy. In fact, for another repetition of the above experiment, we may not obtain an identical favorable permutation or the same optimal λ and L values. So, we repeat the aforementioned grid-searching experiment a number of times (say about 35 times) so that we may obtain a range of λ and a range of L that will be suitable to work with in order to derive the best results out of the proposed framework. We can however, restrict our search for the optimal value of L . Note that the breakdown point of the Median-of-Means estimator is $[(L-1)/2]/n$. In order for the centroid estimates to be stable and not break down in the presence of a little contamination, it is reasonable to assume that we should choose a value of L such that $L = \mathcal{O}(n)$. We choose the median of the clustering accuracies, measured with the Adjusted Rand Index (ARI) (Hubert and Arabie, 1985) so obtained, as a representative of the clustering accuracy of the algorithm. When the ground truth is unavailable, we may use the t-SNE (van der Maaten and Hinton, 2008) plot of the data to form an idea of the ground truth.

3.5 Computational Complexity

In each iteration, our algorithm first ascertains whether an increase in the number of clusters is needed. The centroids are recalculated thereafter, and the cluster assignments are made accordingly. This phase typically takes $\mathcal{O}(nCp)$ time steps to complete, where C represents the number of clusters in that iteration. The calculations presented in the following section assume that the number of clusters is upper bounded by some finite $K < n$. Consequently, the worst case runtime of the DP-MoM algorithm remains $\mathcal{O}(nKp)$ for every iteration.

The computational complexity of the DP-means algorithm is comparable to that of DP-MoM, as each iteration requires $\mathcal{O}(nCp)$ steps to complete, with C denoting the number of clusters in that specific iteration. On the contrary, k -means demands $\mathcal{O}(nkp)$ steps per iteration, with k representing the predefined cluster count. This is typically slated to be lower than that of DP-means or DP-MoM. In the case where we set $k = K$ however, k -means will perform no more efficiently than DP-MoM in terms of runtime.

4 Theoretical Analysis

Let $\mathcal{M}$ denote the set of probability measures P on $\mathbb{R}^p$ with bounded ℓ_2 norm, i.e., for any random vector $\mathbf{X} \sim P$, we have $\mathbb{E}\|\mathbf{X}\|^2 \leq \gamma^2 < \infty$ where $\|\cdot\|$ denotes the ℓ_2 norm. We shall make the standard assumption that all the data points are independent and identically distributed (*i.i.d.*) with finite squared $\mathcal{L}_2$ norm.

A1. $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \stackrel{iid}{\sim} P$ such that $P \in \mathcal{M}$.

We denote the empirical distribution derived from $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$, by P_n , that is, $P_n(A) = \frac{1}{n} \sum_{i=1}^n \mathbf{I}\{\mathbf{X}_i \in A\}$ for any Borel set A . For the sake of notational simplicity, we write $\mu f := \int f d\mu$. The quantities $P_n f_{\Theta}$ and $P f_{\Theta}$ are defined as follows:

$$P_n f_{\Theta} = \frac{1}{n} \sum_{i=1}^n f_{\Theta}(\mathbf{X}_i) \quad \text{and} \quad P f_{\Theta} = \mathbb{E}_P[f_{\Theta}(\mathbf{X})].$$

Further let Θ^* be the global minimizer of $P f_{\Theta}$, and $\hat{\Theta}_n^{(MoM)}$ be the minimizer of (4).

A2. The number of clusters k is bounded above by some finite $K \in \mathbb{N}$, where $K < n$.

The inherent dependency of the number of centroids on the cluster penalty parameter λ makes it possible for us to choose λ appropriately so that the cluster count doesn't exceed K . Moreover, we deduce from A2 that, at a certain juncture, the number of centroids reaches a state of stability. In this state, it is only the cluster centroids themselves that undergo updates during each iteration, while the number of centroids remains constant. This will make our analysis independent of the penalty parameter λ as the term $k\lambda$ is not subject to change after a finite number of iterations. Consequently, beyond a finite number of steps, the objective function effectively reduces to

$$\text{MoM}_L^n(\Theta) := \text{median} \left(\frac{1}{b} \sum_{i \in B_1} f_{\Theta}(\mathbf{X}_i), \frac{1}{b} \sum_{i \in B_2} f_{\Theta}(\mathbf{X}_i), \dots, \frac{1}{b} \sum_{i \in B_\ell} f_{\Theta}(\mathbf{X}_i) \right). \quad (10)$$

Remark 1. In our framework, $|\Psi(\mathbf{x}) - \Psi(\mathbf{y})| \leq \|\mathbf{x} - \mathbf{y}\|_1$.

To see this, recall the definition of Ψ in (7), and note that $|\Psi(\mathbf{x}) - \Psi(\mathbf{y})| = |\min_{1 \leq j \leq K} x_j - \min_{1 \leq j \leq K} y_j|$. Let us assume, without loss of generality, that $\min_{1 \leq j \leq K} x_j \geq \min_{1 \leq j \leq K} y_j$. Hence, $\min_{1 \leq j \leq K} x_j - \min_{1 \leq j \leq K} y_j \leq x_{j^*} - y_{j^*} \leq \|\mathbf{x} - \mathbf{y}\|_1$ where $j^* = \arg\min_{1 \leq j \leq K} y_j$ and the remark is seen to hold.

Since we have chosen an infinite support for P , following Klochkov et al. (2020), we can say that for Θ^* , there must exist a positive real number $M = M(p, k)$ such that $\|\theta_j\|_2 \leq M$ for every $\theta_j \in \Theta^*$ where $j = 1, 2, \dots, k$.

4.1 Analysis under the Median-of-Means (MoM) Paradigm

We represent the set of all inliers as $\{\mathbf{X}_i\}_{i \in \mathcal{I}}$ and the outliers as $\{\mathbf{X}_i\}_{i \in \mathcal{O}}$. We make the following assumptions for determining the rate at which $|P f_{\hat{\Theta}_n^{(MoM)}} - P f_{\Theta^*}|$ approaches 0.

A3. $\{\mathbf{X}_i\}_{i \in \mathcal{I}} \sim P$ are i.i.d. with $P \in \mathcal{M}$.

A4. $\exists \eta > 0$ such that $L > (2 + \eta)|\mathcal{O}|$.

Assumption A3 ensures that the inliers arise independently from some distribution P . A4 guarantees that at least half of the L partitions are devoid of outliers. Such an assumption involving an upper bound on the number of outliers is essential as a high degree of contamination would imply that these so called 'outliers' need to be treated as the data. This is, in fact, a milder requirement compared to the condition $L > 4|\mathcal{O}|$ imposed in the recent work (Lecu   et al., 2020). Crucially, we

highlight that no distributional assumptions are imposed on the outliers, permitting them to be unbounded, originate from heavy-tailed distributions, or exhibit any dependence structure among themselves.

Since $\|\boldsymbol{\theta}\|_2 \leq M$ for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}^*$, the search space for $\hat{\boldsymbol{\Theta}}_n^{(\text{MoM})}$ may be constrained to $\mathcal{G}$. Subsequently, we establish a high probability bound on $\sup_{\boldsymbol{\Theta} \in \mathcal{G}} |\text{MoM}_L^n(f_{\boldsymbol{\Theta}}) - Pf_{\boldsymbol{\Theta}}|$, and further use it to bound $|Pf_{\hat{\boldsymbol{\Theta}}_n^{(\text{MoM})}} - Pf_{\boldsymbol{\Theta}^*}|$. We use “ $\lesssim$ ” to denote the fact that a quantity is lesser than a constant multiple of the other.

Theorem 4.1. *Under A3-A4, with probability at least $1 - 2e^{-2L\delta^2}$,*

$$\sup_{\boldsymbol{\Theta} \in \mathcal{G}} |\text{MoM}_L^n(f_{\boldsymbol{\Theta}}) - Pf_{\boldsymbol{\Theta}}| \lesssim M(M + 2\gamma)n^{-1/2}.$$

We present below a corollary that aids us in controlling the absolute difference $|Pf_{\hat{\boldsymbol{\Theta}}_n^{(\text{MoM})}} - Pf_{\boldsymbol{\Theta}^*}|$.

Corollary 4.1. *Under A3-A4, with probability at least $1 - 2e^{-2L\delta^2}$,*

$$\left| Pf_{\hat{\boldsymbol{\Theta}}_n^{\text{MoM}}} - Pf_{\boldsymbol{\Theta}^*} \right| \lesssim M(M + 2\gamma)n^{-1/2}.$$

4.2 Asymptotic Properties: Consistency and Rate of Convergence

We now consider the classical setting where p is held constant, and demonstrate that the previously presented results imply consistency, with rate of convergence of the order of $\mathcal{O}(n^{-1/2})$. We first follow the same idea of convergence of $\hat{\boldsymbol{\Theta}}_n^{\text{MoM}}$ to $\boldsymbol{\Theta}^*$ that is outlined in Pollard (1981). Since the centroids are unique up to rearrangement of labels, our concept of dissimilarity

$$\mathcal{D}(\boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2) = \min_{H \in \mathcal{P}_K} \|\boldsymbol{\Theta}_1 - H\boldsymbol{\Theta}_2\|_F$$

is considered over $\mathcal{P}_K$ the set of all real permutation matrices of order K , where $\|\cdot\|_F$ represents the Frobenius norm. The sequence $\boldsymbol{\Theta}_n \rightarrow \boldsymbol{\Theta}$ if $\lim_{n \rightarrow \infty} \mathcal{D}(\boldsymbol{\Theta}_n, \boldsymbol{\Theta}) = 0$. Following Terada (2014); Chakraborty et al. (2020), we assume the identifiability condition:

A5. $\forall \eta > 0, \exists \varepsilon > 0$, such that $\mathcal{D}(\boldsymbol{\Theta}, \boldsymbol{\Theta}^*) > \eta \implies Pf_{\boldsymbol{\Theta}} > Pf_{\boldsymbol{\Theta}^*} + \varepsilon$.

We now examine the consistency of $\hat{\boldsymbol{\Theta}}_n^{\text{MoM}}$. In addition, we analyze the rate at which $|Pf_{\hat{\boldsymbol{\Theta}}_n^{\text{MoM}}} - Pf_{\boldsymbol{\Theta}^*}|$ approaches 0. Theorem 4.2 affirms that strong consistency holds, with $\mathcal{O}(n^{-1/2})$ rate of convergence. Before delving further, recall that $X_n = \mathcal{O}_P(a_n)$ if the sequence X_n/a_n is bounded in probability.

A6. *The number of buckets $L \rightarrow \infty$ as $n \rightarrow \infty$.*

Theorem 4.2. *Under A1-A5,*

$$\left| Pf_{\hat{\boldsymbol{\Theta}}_n^{\text{MoM}}} - Pf_{\boldsymbol{\Theta}^*} \right| = \mathcal{O}_P(n^{-1/2})$$

and consequently $Pf_{\hat{\boldsymbol{\Theta}}_n^{\text{MoM}}} \xrightarrow{p} Pf_{\boldsymbol{\Theta}^*}$. Furthermore, under A6, we have $\hat{\boldsymbol{\Theta}}_n^{\text{MoM}} \xrightarrow{p} \boldsymbol{\Theta}^*$.

5 Experiments

We now empirically compare our proposed framework with existing clustering approaches to thoroughly validate and evaluate its effectiveness. The accuracy of cluster assignments has been rigorously assessed using the Adjusted Rand Index (ARI), a robust measure of clustering performance. Our evaluation encompasses an extensive array of competing centroid-based clustering methods, including renowned techniques such as k -means++, Sparse k -means (SKM) (Witten and Tibshirani, 2010), k -medians, Partition around Medoids (PAM) (der Laan et al., 2003), Robust Continuous Clustering (RCC) (Shah and Koltun, 2017), DP-means (Kulis and Jordan, 2012), k -bootstrap Median-of-Means (K -bMOM) (Brunet-Saumard et al., 2022), Median-of-Means with Power k -means (MoMPKM), and Ordered Weighted l_1 k -means (OWL k -means) (Chakraborty et al., 2023). These state-of-the-art algorithms are benchmarked against the proposed DP-MoM algorithm across various experimental scenarios. The simulation experiments were conducted using a computer equipped with Intel(R) Core(TM)i3-7020U 2.30GHz processor, 4GB RAM, 64-bit Windows 10 operating system in the R programming language (R Core Team, 2022). The relevant codes can be accessed at <https://github.com/jyotishkarc/Clustering-Techniques/tree/main/DP-MoM>.

Our experiments involve implementing the aforementioned techniques on a simulated dataset as well as several datasets from the UCI Machine Learning Repository¹ and the CompCancer database². Owing to the fact that our clustering technique relies on randomization while partitioning the data into buckets, the accuracy measure has been computed as the median value of the obtained ARI over 30 test runs. It was observed, for most of the datasets, that DP-MoM performed considerably better than its competitors in terms of ARI. Apart from this, two other experiments were conducted to assess the strength of the algorithm in terms of robustness and ability to detect clusters of various shapes that were in proximity to one another in terms of their pairwise Euclidean distances.

5.1 Simulation Studies

Study of Robustness on Simulated Data: 30 data points are generated from each of the 4 quadrants in the 2-dimensional Euclidean plane using a special generation scheme. For the first quadrant, we generate $R_i \sim U(0, 1)$ and $\theta_i \sim U\left(\frac{\pi}{36}, \frac{17\pi}{36}\right)$. Once this is done for all $i = 1, 2, \dots, 30$, we set $X_i = (R_i \cos \theta_i, R_i \sin \theta_i)$ for all $i = 1, 2, \dots, 30$ as our data points in the first quadrant. In the other quadrants, we draw R_i in the same way and generate θ_i uniformly from $\left(\frac{(j-1)\pi}{2} + \frac{\pi}{36}, \frac{j\pi}{2} - \frac{\pi}{36}\right)$ for the j^{th} quadrant. We place the data points lying in the same quadrant in the same cluster. Just like in the experiment using the *Jain* dataset, outliers have been generated uniformly on $[-1, 1] \times [-1, 1]$. 15, 15, and 20 outliers were introduced in three stages, respectively, so that the total number of data points stood at 135, 150, and 170 respectively. While the efficiency of the other competing algorithms plummeted or showed erratic behavior (often combined with low clustering accuracy), the ARI corresponding to DP-MoM did not waver. Figure 2 depicts the superiority of our proposed algorithm over the other existing clustering techniques.

¹<https://archive.ics.uci.edu/>

²<https://schlieplab.org/Static/Supplements/CompCancer/>

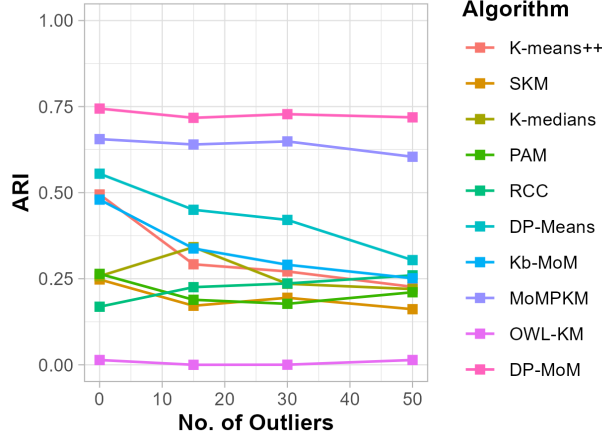


Figure 2: Line plots of ARI produced by different algorithms on simulated datasets, for increasingly higher number of outliers. DP-MoM is observed to perform uniformly better than all the competing methods.

5.2 Real Data Experiments

5.2.1 Introducing Outliers - A Case Study

We have picked the dataset *Jain* (Jain and Law, 2005) for this experiment. *Jain* is a 2-dimensional dataset with 373 data points. The 2 natural clusters are shaped like boomerangs, as can be seen in Figure 1 in Section 1. The performance of the algorithms was assessed on the original dataset. Afterward, several outliers were uniformly generated throughout the range of the data. 20 fresh outliers were introduced in each of 4 stages and at each stage, the algorithms were pitted against each other again. Even with the introduction of 80 outliers, DP-MoM remained remarkably robust, consistently achieving a clustering accuracy of nearly 0.9 in terms of ARI (while the maximum ARI achieved was above that figure in all but one stage). Conversely, many other competing algorithms struggled to maintain their performance in the face of increasing outlier counts. They exhibited significant fluctuations in ARI as the number of outliers rose. Even the ones that maintained stability could only muster a measly ARI of 0.42, as did all the other competing techniques.

5.2.2 Further Experiments

For a comprehensive performance evaluation of our proposed clustering algorithm in situations where the underlying data distributions are unknown, we implement DP-MoM on several real datasets from the Compcancer database and the UCI Repository. Additionally, we implement some state-of-the-art clustering algorithms mentioned at the start of this section, on the same datasets, and compare the corresponding ARI values against that of DP-MoM. Since DP-MoM is a randomized algorithm in the sense that its cluster assignment is dependent on the initial dataset partitioning into buckets, we implement DP-MoM on each dataset 30 times independently, and report the median ARI. The same procedure is followed while reporting the ARI values for the competing algorithms.

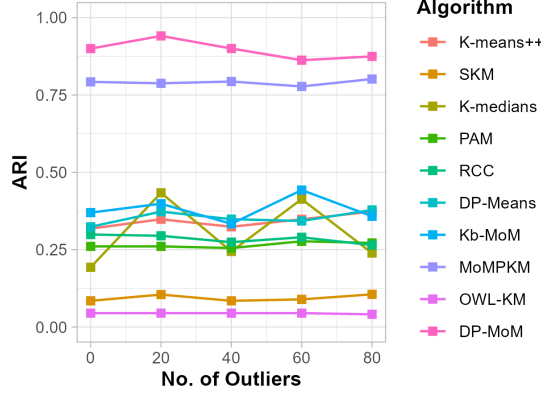


Figure 3: Line plots of ARI produced by different algorithms, for increasingly higher number of noisy observations introduced in the *Jain* dataset. DP-MoM performs better than all the competing methods.

Table 1: ARI values corresponding to clustering via state-of-the-art algorithms as well as DP-MoM on UCI datasets

Dataset	Description			State-of-the-Art Algorithms									DP-MoM
	<i>n</i>	<i>p</i>	<i>K</i>	KM++	SKM	KMed	PAM	RCC	DPM	KbMoM	MoMPKM	OWL-KM	
Iris	150	4	3	0.7237	0.7960	0.7515	0.6325	0.8090	0.7515	0.7565	0.8647	0.6339	0.9799
Glass	214	9	7	0.3728	0.3595	0.3367	0.3501	0.3930	0.4472	0.3467	0.2484	0.2659	0.3190
WDBC	569	30	2	0.4223	0.4223	0.4603	0.4587	0.4146	0.4479	0.4560	0.6839	0.5897	0.6798
E.Coli	336	7	8	0.5001	0.4918	0.5346	0.5407	0.5350	0.6663	0.6216	0.4952	0.2174	0.7835
Wine	178	13	3	0.4140	0.4287	0.4226	0.4189	0.3564	0.4094	0.4227	0.5518	0.3470	0.5820
Thyroid	215	5	3	0.3936	0.2145	0.1450	0.2144	0.5186	0.4971	0.3032	0.5995	0.4392	0.8842
Zoo	101	16	7	0.7376	0.7516	0.6730	0.6566	0.7173	0.8270	0.4978	0.7603	0.8408	0.8477
soybean	47	35	4	0.7143	0.7138	0.7108	0.7437	0.8268	0.7368	0.82678	0.7417	0.5452	0.9533
Average Rank				6.5625	6.1875	6.9375	6.5000	5.1875	4.6875	5.4375	4.2500	7.2500	2.0000

Table 2: ARI values corresponding to clustering via state-of-the-art algorithms as well as DP-MoM on Compccancer datasets

Dataset	Description			State-of-the-Art Algorithms									DP-MoM
	<i>n</i>	<i>p</i>	<i>K</i>	KM++	SKM	KMed	PAM	RCC	DPM	KbMoM	MoMPKM	OWL-KM	
golub_1999_v2	72	1868	3	0.4334	0.6876	0.6116	0.7716	0.0000	0.6421	0.5664	0.6361	0.7438	0.7798
west_2001	49	1198	2	0.1527	0.0002	0.0886	0.1058	0.0000	0.1715	0.3061	0.4761	0.5613	0.5035
pomeroy_2002_v2	42	857	5	0.0000	0.4924	0.0000	0.0000	0.0000	0.0514	0.3583	0.4685	0.5175	0.5446
singh_2002	102	339	2	0.0259	0.0330	0.0259	0.0330	0.0000	0.0574	0.0330	0.3433	0.0483	0.8135
tomlins_v2	92	1288	4	0.1418	0.1245	0.0100	0.2134	0.0000	0.1814	0.1730	0.2775	0.1993	0.3985
alizadeh_2000_v1	42	1095	2	0.0127	0.0000	0.0000	0.2564	0.0000	0.0023	0.0613	0.2714	0.0889	0.3716
armstrong_2002_v2	72	2194	3	0.5123	0.5448	0.6625	0.4584	0.0000	0.4660	0.4992	0.6365	0.9186	0.8332
bredel_2005	50	832	3	0.2000	0.3525	0.0098	0.4760	0.0000	0.0893	0.2315	0.4877	0.2996	0.5841
Average Rank				7.2500	6.0000	7.7500	5.3125	9.6875	5.8750	5.7500	3.1250	3.0000	1.2500

5.2.3 Friedman’s Rank Test

Friedman’s rank test (Friedman, 1937) is employed to discern whether a significant difference exists in the performance of the algorithms applied to our datasets. This assessment unfolds across three stages. In the initial stage, the test encompasses all clustering algorithms under consideration. Moving to the second stage, the analysis omits DP-MoM while incorporating the other algorithms. In the third stage, both MoMPKM and DP-MoM are excluded from the test. The calculated p -values for these three stages are as follows: 1.57×10^{-7} , 0.0021, and 0.0599 respectively. The null hypothesis, which posits no significant variance in clustering accuracy among the tested algorithms, is rejected in the first and second stages. However, it is accepted in the third stage. This outcome underscores that MoMPKM and DP-MoM emerge as the most proficient clustering algorithms at our disposal. Further assessments indicate that MoMPKM is outperformed comprehensively by the novel DP-MoM.

5.2.4 Sign Test and Wilcoxon Signed Rank (WSR) Test

We also perform the *Sign Test* and *Wilcoxon Signed Rank Test* to compare our proposed algorithm individually with every other competing algorithm mentioned in Tables 1 and 2 and check whether DP-MoM performs significantly better than each of the aforementioned state-of-the-art algorithms. It is evident from the p -values that the null hypotheses: H_{0s} : The said algorithm is better than our proposed algorithm DP-MoM (*Sign Test*) and H_{0w} : The said algorithm is equivalent to our proposed framework DP-MoM (*Wilcoxon’s signed Rank Test*) are rejected in favor of the alternative H_1 : DP-MoM performs significantly better than the other state-of-the-art clustering algorithm in question for a test with level of significance 0.01. In a majority of the cases, our proposed DP-MoM algorithm is the standout performer. However, in the 3 cases where its performance is slightly suboptimal with respect to that of MoMPKM and OWL k -means, the results of the statistical tests presented in Table 3, indicate that this drop in performance is not statistically significant at the specified level.

Table 3: Summary of the Statistical Test Results for level of significance 0.01

Clustering Algorithm	Sign Test		WSR Test	
	Statistic	p -value	Statistic	p -value
k -means ++	15	0.0002594	135	0.0000305
Sparse k -means	15	0.0002594	135	0.0000305
k -medians	15	0.0002594	135	0.0000305
Partition Around Medoids	15	0.0002594	134	0.0000458
Robust Continuous Clustering	15	0.0002594	135	0.0000305
DP means	15	0.0002594	133	0.0000763
Kb MoM	15	0.0002594	135	0.0000305
MoM Power k -means	15	0.0002594	135	0.0000305
OWL k -means	14	0.0020900	125	0.0008392

Tables 4 and 5 provide the range of the penalty parameter λ that enables us to cluster each dataset more efficiently. The predicted number of clusters are also displayed. Note that the number

of clusters have been calculated after assigning the data points in the clusters containing less than 3 observations, to the nearest cluster containing at least 3 observations.

Table 4: Range of optimal λ and estimated number of clusters for implementing DP-MoM on UCI datasets

Dataset	Range of Optimal λ	Estimated Clusters
Iris	5.129 - 6.535	3
Glass	4.207 - 18.841	6
WDBC	2.5×10^6 - 6×10^6	2
E. Coli	0.3008 - 0.3571	8
Wine	3.78×10^5 - 3.93×10^5	2
Thyroid	1769 - 2123	5
Zoo	7.048 - 10.360	6
soybean	18.59 - 25.70	4

Table 5: Range of optimal λ and estimated number of clusters for implementing DP-MoM on Compcancer datasets

Dataset	Range of Optimal λ	Estimated Clusters
golub_1999_v2	2.48×10^9 - 3.05×10^9	3
west_2001	1.77×10^9 - 3.11×10^9	2
pomeroy_2002_v2	2.94×10^9 - 4.70×10^9	4
singh_2002	0.14×10^9 - 0.17×10^9	2
tomlins_v2	175.6 - 308.0	2
alizadeh_2000_v1	762.8 - 780.0	2
armstrong_2002_v2	5.85×10^9 - 7.08×10^9	3
breidel_2005	672.4 - 978.8	2

6 Discussion

In this article, we proposed a new clustering algorithm that integrates two major clustering paradigms, viz., centroid-based and model-based clustering, and is intended to perform well on noisy or outlier-laden data. We utilize the Median-of-Means (MoM) estimator to deal with noise or outliers present in data, and a Bayesian nonparametric modelling ensures that the number of clusters need not be specified. Unlike conventional clustering methods, which tackle only one of these challenges, our proposed algorithm adeptly tackles both at the same time. Following our comprehensive theoretical analysis of error rate bounds, augmented by extensive simulation studies and real-world data analysis, we showcase the superiority of our method against quite a few prominent clustering algorithms.

However, our proposed technique is not without its shortcomings; the parameter λ is notoriously hard to tune without resorting to grid searching using clustering efficiency. On the bright side, our

technique works quite well even in the high-dimensional setting, and it might be fruitful to explore convergence results in this regime to further enhance its applicability and efficacy in the future.

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Supplementary Material

A Proofs of Results in Section 4

A.1 Proof of Theorem 4.1

Proof. We represent the empirical distribution of $\{\mathbf{X}_i\}_{i \in B_\ell}$ by P_{B_ℓ} . Fix $\varepsilon > 0$. We will first establish a concentration inequality on $\sup_{\boldsymbol{\theta} \in \mathcal{G}} |\text{MoM}_L^n(f_{\boldsymbol{\theta}}) - Pf_{\boldsymbol{\theta}}|$ where $\mathcal{G}$ denotes the collection of all $\boldsymbol{\theta}$ such that $\|\boldsymbol{\theta}\| \leq M$ for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$. To that end, we bound the probabilities of the events $\{\sup_{\boldsymbol{\theta} \in \mathcal{G}} (\text{MoM}_L^n(f_{\boldsymbol{\theta}}) - Pf_{\boldsymbol{\theta}}) > \varepsilon\}$ and $\{\sup_{\boldsymbol{\theta} \in \mathcal{G}} (Pf_{\boldsymbol{\theta}} - \text{MoM}_L^n(f_{\boldsymbol{\theta}})) > \varepsilon\}$ individually. Note that

$$\sup_{\boldsymbol{\theta} \in \mathcal{G}} \sum_{\ell=1}^L \mathbb{1}\{(P - P_{B_\ell})f_{\boldsymbol{\theta}} > \varepsilon\} > \frac{L}{2} \implies \sup_{\boldsymbol{\theta} \in \mathcal{G}} (Pf_{\boldsymbol{\theta}} - \text{MoM}_L^n(f_{\boldsymbol{\theta}})) > \varepsilon. \quad (11)$$

To see this, suppose on the contrary that

$$\sup_{\boldsymbol{\theta} \in \mathcal{G}} (Pf_{\boldsymbol{\theta}} - \text{MoM}_L^n(f_{\boldsymbol{\theta}})) \leq \varepsilon$$

but

$$\sup_{\boldsymbol{\theta} \in \mathcal{G}} \sum_{\ell=1}^L \mathbb{1}\{(P - P_{B_\ell})f_{\boldsymbol{\theta}} > \varepsilon\} > \frac{L}{2}$$

. Then for all $\boldsymbol{\theta} \in \mathcal{G}$, we must have

$$(Pf_{\boldsymbol{\theta}} - \text{MoM}_L^n(f_{\boldsymbol{\theta}})) \leq \varepsilon$$

which implies that for all $\boldsymbol{\theta} \in \mathcal{G}$

$$\sum_{\ell=1}^L \mathbb{1}\{(P - P_{B_\ell})f_{\boldsymbol{\theta}} \leq \varepsilon\} \geq \frac{L}{2} \implies \sum_{\ell=1}^L \mathbb{1}\{(P - P_{B_\ell})f_{\boldsymbol{\theta}} > \varepsilon\} \leq \frac{L}{2}$$

which in turn implies that

$$\sup_{\boldsymbol{\theta} \in \mathcal{G}} \sum_{\ell=1}^L \mathbb{1}\{(P - P_{B_\ell})f_{\boldsymbol{\theta}} > \varepsilon\} \leq \frac{L}{2}$$

which is a contradiction. Let $\varphi(t) = (t-1)\mathbb{1}\{1 \leq t \leq 2\} + \mathbb{1}\{t > 2\}$ where $\mathbb{1}\{\cdot\}$ is the indicator function. Evidently,

$$\mathbb{1}\{t \geq 2\} \leq \varphi(t) \leq \mathbb{1}\{t \geq 1\}. \quad (12)$$

We observe that

$$\begin{aligned} \sup_{\boldsymbol{\theta} \in \mathcal{G}} \sum_{\ell=1}^L \mathbb{1}\{(P - P_{B_\ell})f_{\boldsymbol{\theta}} > \varepsilon\} &\leq \sup_{\boldsymbol{\theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \mathbb{E} \varphi \left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\theta}}}{\varepsilon} \right) + |\mathcal{O}| \\ &\quad + \sup_{\boldsymbol{\theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \left[\varphi \left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\theta}}}{\varepsilon} \right) - \mathbb{E} \varphi \left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\theta}}}{\varepsilon} \right) \right]. \end{aligned} \quad (13)$$

Towards bounding $\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell=1}^L \mathbb{1} \{(P - P_{B_\ell})f_{\boldsymbol{\Theta}} > \varepsilon\}$, we first bound $\mathbb{E}\varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right)$. Note that

$$\begin{aligned} \mathbb{E}\varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) &\leq \mathbb{P}\left[(P - P_{B_\ell})f_{\boldsymbol{\Theta}} > \frac{\varepsilon}{2}\right] \leq \frac{4}{\varepsilon^2} \text{Var} P_{B_\ell} f_{\boldsymbol{\Theta}} \\ &= \frac{4L}{|\mathcal{I}|\varepsilon^2} \mathbb{E} \min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} (\|\boldsymbol{\theta}\|^2 - 2\boldsymbol{\theta}^\top \mathbf{X})^2 \\ &\leq \frac{4L}{|\mathcal{I}|\varepsilon^2} M^2 (M + 2\gamma)^2. \end{aligned}$$

The last inequality follows by Chebyshev's inequality after observing that $\mathbb{E}P_{B_\ell}f_{\boldsymbol{\Theta}} = Pf_{\boldsymbol{\Theta}}$. We now turn to bounding the term

$$\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \left[\varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) - \mathbb{E}\varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) \right].$$

Appealing to Theorem 26.5 of [Shalev-Shwartz and Ben-David \(2014\)](#), for all $\boldsymbol{\Theta} \in \mathcal{G}$, we have

$$\begin{aligned} &\frac{1}{L} \sum_{\ell \in \mathcal{L}} \varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) - \mathbb{E} \left[\frac{1}{L} \sum_{\ell \in \mathcal{L}} \varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) \right] \\ &\leq 2\mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \frac{1}{L} \sum_{\ell \in \mathcal{L}} \sigma_\ell \varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) \right] + \delta. \end{aligned} \quad (14)$$

with probability at least $1 - e^{-2L\delta^2}$, where $\{\sigma_\ell\}_{\ell \in \mathcal{L}}$ are i.i.d. Rademacher random variables independent of the sample $\mathcal{X}$. Consider i.i.d. Rademacher random variables $\{\xi_i\}_{i=1}^n$ independent of $\{\sigma_\ell\}_{\ell \in \mathcal{L}}$. As a consequence of equation (14), we get,

$$\begin{aligned} &\frac{1}{L} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \left[\varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) - \mathbb{E}\varphi\left(\frac{2(P - P_{B_\ell})f_{\boldsymbol{\Theta}}}{\varepsilon}\right) \right] \\ &\leq \frac{4}{L\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \sigma_\ell (P - P_{B_\ell})f_{\boldsymbol{\Theta}} \right] + \delta. \end{aligned} \quad (15)$$

Equation (15) follows from the fact that $\varphi(\cdot)$ is 1-Lipschitz and Lemma 26.9 of [Shalev-Shwartz and Ben-David \(2014\)](#). We now consider random variables $\mathcal{X}' = \{\mathbf{X}'_1, \dots, \mathbf{X}'_n\}$, which are i.i.d. and follow the law P . Equation (15) further yields

$$\begin{aligned} &= \frac{4}{L\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \sigma_\ell \mathbb{E}_{\mathcal{X}'} ((P'_{B_\ell} - P_{B_\ell})f_{\boldsymbol{\Theta}}) \right] + \delta \\ &\leq \frac{4}{L\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \sigma_\ell (P'_{B_\ell} - P_{B_\ell})f_{\boldsymbol{\Theta}} \right] + \delta \end{aligned}$$

This inequality follows by employing Jensen's inequality as supremum is a convex function, and the expectation in the second step is taken with respect to both the original sample $\mathcal{X}$ as well as the ghost sample $\mathcal{X}'$ and the Rademacher random variables $\{\sigma_\ell\}_{\ell \geq 1}$.

$$= \frac{4}{L\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \sigma_\ell \frac{1}{b} \sum_{i \in B_\ell} (f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) - f_{\boldsymbol{\Theta}}(\mathbf{X}_i)) \right] + \delta$$

$$= \frac{4}{bL\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \sigma_\ell \sum_{i \in B_\ell} \xi_i (f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) - f_{\boldsymbol{\Theta}}(\mathbf{X}_i)) \right] + \delta \quad (16)$$

$$= \frac{4}{|\mathcal{I}|\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell \in \mathcal{L}} \sum_{i \in B_\ell} \sigma_\ell \xi_i (f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) - f_{\boldsymbol{\Theta}}(\mathbf{X}_i)) \right] + \delta$$

$$= \frac{4}{|\mathcal{I}|\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{i \in \mathcal{J}} \gamma_i (f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) - f_{\boldsymbol{\Theta}}(\mathbf{X}_i)) \right] + \delta \quad (17)$$

$$\leq \frac{4}{|\mathcal{I}|\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{i \in \mathcal{J}} \gamma_i f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) + \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{i \in \mathcal{J}} \gamma_i f_{\boldsymbol{\Theta}}(\mathbf{X}_i) \right] + \delta \quad (18)$$

$$\leq \frac{4}{|\mathcal{I}|\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{i \in \mathcal{J}} \gamma_i f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) \right] + \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{i \in \mathcal{J}} \gamma_i f_{\boldsymbol{\Theta}}(\mathbf{X}_i) \right] + \delta \quad (19)$$

$$= \frac{8}{\varepsilon} \mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{J}} \gamma_i f_{\boldsymbol{\Theta}}(\mathbf{X}_i) \right] + \delta. \quad (20)$$

Here $\mathcal{J}$ is the set of observations in the partitions not containing an outlier. Equation (16) follows from the fact that $(f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) - f_{\boldsymbol{\Theta}}(\mathbf{X}_i)) \stackrel{d}{=} \xi_i (f_{\boldsymbol{\Theta}}(\mathbf{X}'_i) - f_{\boldsymbol{\Theta}}(\mathbf{X}_i))$ as $\{\xi_i\}_{i \geq 1}$ are Rademacher random variables independent of the sample $\mathcal{X}$. In equation (17), $\{\gamma_i\}_{i \in \mathcal{J}}$ are independent Rademacher random variables since the product of two independent Rademacher random variables is also a Rademacher random variable.

We compute the Rademacher complexity in Eq (20) as follows: Note that $|\Psi(\mathbf{x}) - \Psi(\mathbf{y})| \leq \|\mathbf{x} - \mathbf{y}\|_2$. To see this, observe that $|\Psi(\mathbf{x}) - \Psi(\mathbf{y})| \leq |\min_{1 \leq j \leq k} x_j - \min_{1 \leq i \leq k} y_i|$. Now, without loss of generality, let $\min_{1 \leq j \leq k} x_j \geq \min_{1 \leq i \leq k} y_i$. Then $|\Psi(\mathbf{x}) - \Psi(\mathbf{y})| \leq x_{j^*} - y_{j^*} \leq \|\mathbf{x} - \mathbf{y}\|_2$ where $j^* = \operatorname{argmin}_{1 \leq i \leq k} y_i$. Let $g_{\boldsymbol{\Theta}}(\mathbf{x}) = (-2\langle \mathbf{x}, \boldsymbol{\theta}_1 \rangle + \|\boldsymbol{\theta}_1\|^2, \dots, -2\langle \mathbf{x}, \boldsymbol{\theta}_k \rangle + \|\boldsymbol{\theta}_k\|^2)$, we have

$$|f_{\boldsymbol{\Theta}}(\mathbf{X}_i) - f_{\boldsymbol{\Theta}}(\mathbf{X}_i)| \leq \|g_{\boldsymbol{\Theta}}(\mathbf{X}_i) - g_{\boldsymbol{\Theta}}(\mathbf{X}_i)\|_2.$$

Further Maurer's vector contraction inequality yields

$$\mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{J}} \gamma_i f_{\boldsymbol{\Theta}}(\mathbf{X}_i) \right] \leq \frac{\sqrt{2}}{|\mathcal{I}|} \left(2\mathbb{E} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{j=1}^k \sum_{i=1}^{|\mathcal{I}|} \varepsilon_{i,j} \langle \mathbf{X}_i, \boldsymbol{\theta}_j \rangle + \mathbb{E} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{j=1}^k \sum_{i=1}^{|\mathcal{I}|} \varepsilon_{i,j} \|\boldsymbol{\theta}_j\|^2 \right).$$

Moreover, by Khintchine's Inequality, we have

$$\begin{aligned} \mathbb{E} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{j=1}^k \sum_{i=1}^{|\mathcal{I}|} \varepsilon_{i,j} \langle \mathbf{X}_i, \boldsymbol{\theta}_j \rangle &\leq \sum_{j=1}^k \mathbb{E} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \left\langle \sum_{i=1}^{|\mathcal{I}|} \varepsilon_{i,j} \mathbf{X}_i, \boldsymbol{\theta}_j \right\rangle \\ &\leq \sum_{j=1}^k \mathbb{E} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \left\| \sum_{i=1}^{|\mathcal{I}|} \varepsilon_{i,j} \mathbf{X}_i \right\| \|\boldsymbol{\theta}_j\| \\ &\leq kM \max_{j \leq k} \mathbb{E} \left\| \sum_{i=1}^{|\mathcal{I}|} \varepsilon_{i,j} \mathbf{X}_i \right\| \end{aligned}$$

$$\begin{aligned}
&\leq kM \max_{j \leq k} \sqrt{\mathbb{E} \left\| \sum_{i=1}^{|I|} \varepsilon_{i,j} \mathbf{X}_i \right\|^2} \\
&\leq kM \sqrt{\sum_{i=1}^{|I|} \mathbb{E} \|\mathbf{X}_i\|^2} \leq kM\gamma\sqrt{|I|}
\end{aligned}$$

and

$$\mathbb{E} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{j=1}^k \sum_{i=1}^{|I|} \varepsilon_{i,j} \|\boldsymbol{\theta}_j\|^2 \leq \sum_{j=1}^k \mathbb{E} \sup_{\boldsymbol{\Theta} \in \mathcal{G}} \left\| \sum_{i=1}^{|I|} \varepsilon_{i,j} \right\| \|\boldsymbol{\theta}_j\|^2 \leq kM^2 \sqrt{|I|}.$$

Thus, we have

$$\mathbb{E} \left[\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \frac{1}{|I|} \sum_{i \in \mathcal{I}} \gamma_i f_{\boldsymbol{\Theta}}(\mathbf{X}_i) \right] \leq \frac{\sqrt{2}kM(M+2\gamma)}{\sqrt{|I|}}.$$

From the above computations, we conclude that, with probability of at least $1 - e^{-2L\delta^2}$,

$$\sup_{\boldsymbol{\Theta} \in \mathcal{G}} \sum_{\ell=1}^L \mathbb{1} \{(P - P_{B_\ell})f_{\boldsymbol{\Theta}} > \varepsilon\} \leq L \left(\frac{4}{|I|\varepsilon^2} M^2(M+2\gamma)^2 + \frac{|\mathcal{O}|}{L} + \frac{8\sqrt{2}kM(M+2\gamma)}{\varepsilon\sqrt{|I|}} + \delta \right). \quad (21)$$

We choose $\delta = \frac{2}{4+\eta} - \frac{|\mathcal{O}|}{L}$, and

$$\varepsilon = \max \left\{ \frac{4\sqrt{(\eta+4)}}{\sqrt{\eta}\sqrt{n}} M(M+2\gamma), \frac{32(\eta+4)}{\eta\sqrt{n}} M(M+2\gamma) \right\}.$$

This makes the right hand side of (21) strictly smaller than $\frac{L}{2}$. Thus, we have shown that

$$\mathbb{P} \left(\sup_{\boldsymbol{\Theta} \in \mathcal{G}} (Pf_{\boldsymbol{\Theta}} - \text{MoM}_L^n(f_{\boldsymbol{\Theta}})) > \varepsilon \right) \leq e^{-2L\delta^2}.$$

By a similar argument, it follows that,

$$\mathbb{P} \left(\sup_{\boldsymbol{\Theta} \in \mathcal{G}} (\text{MoM}_L^n(f_{\boldsymbol{\Theta}}) - Pf_{\boldsymbol{\Theta}}) > \varepsilon \right) \leq e^{-2L\delta^2}.$$

From the two aforementioned inequalities, we obtain,

$$\mathbb{P} \left(\sup_{\boldsymbol{\Theta} \in \mathcal{G}} |\text{MoM}_L^n(f_{\boldsymbol{\Theta}}) - Pf_{\boldsymbol{\Theta}}| > \varepsilon \right) \leq 2e^{-2L\delta^2}.$$

i.e., with at least probability $1 - 2e^{-2L\delta^2}$,

$$\begin{aligned}
&\sup_{\boldsymbol{\Theta} \in \mathcal{G}} |\text{MoM}_L^n(f_{\boldsymbol{\Theta}}) - Pf_{\boldsymbol{\Theta}}| \\
&\leq \max \left\{ \frac{4\sqrt{(\eta+4)}}{\sqrt{\eta}\sqrt{n}} M(M+2\gamma), \frac{32(\eta+4)}{\eta\sqrt{n}} M(M+2\gamma) \right\} \lesssim \frac{M(M+2\gamma)}{\sqrt{n}}.
\end{aligned}$$

□

A.2 Proof of Corollary 4.1

Proof. Note that the following holds with probability at least $1 - 2e^{-2L\delta^2}$:

$$\begin{aligned}
& \left| Pf_{\hat{\Theta}_n^{\text{MoM}}} - Pf_{\Theta^*} \right| \\
&= Pf_{\hat{\Theta}_n^{\text{MoM}}} - Pf_{\Theta^*} \\
&= Pf_{\hat{\Theta}_n^{\text{MoM}}} - \text{MoM}_L^n(f_{\hat{\Theta}_n^{\text{MoM}}}) + \text{MoM}_L^n(f_{\hat{\Theta}_n^{\text{MoM}}}) - \text{MoM}_L^n(f_{\Theta^*}) + \text{MoM}_L^n(f_{\Theta^*}) - Pf_{\Theta^*} \\
&\leq Pf_{\hat{\Theta}_n^{\text{MoM}}} - \text{MoM}_L^n(f_{\hat{\Theta}_n^{\text{MoM}}}) + \text{MoM}_L^n(f_{\Theta^*}) - Pf_{\Theta^*} \\
&\leq 2 \sup_{\Theta \in \mathcal{G}} |\text{MoM}_L^n(f_{\Theta}) - Pf_{\Theta}| \lesssim M(M + 2\gamma)n^{-1/2}
\end{aligned}$$

Note that the penultimate inequality follows by the definition of $\hat{\Theta}_n^{\text{MoM}}$. $\square$

A.3 Proof of Corollary 4.2

Proof. Since $\delta = 2/(4 + \eta) - |\mathcal{O}|/L > \eta/(4 + \eta)(2 + \eta)$ which implies that $e^{-2L\delta^2} = o(1)$ as $n \rightarrow \infty$. Thus,

$$\mathbb{P} \left(\left| Pf_{\hat{\Theta}_n^{\text{MoM}}} - Pf_{\Theta^*} \right| = \mathcal{O}(n^{-1/2}) \right) \geq 1 - o(1)$$

which implies that $\left| Pf_{\hat{\Theta}_n^{\text{MoM}}} - Pf_{\Theta^*} \right| = \mathcal{O}_P(n^{-1/2})$.

Further, as $n \rightarrow \infty$, we must have $\left| Pf_{\hat{\Theta}_n^{\text{MoM}}} - Pf_{\Theta^*} \right| = o_P(n^{-1/2})$ which in turn implies that $Pf_{\hat{\Theta}_n^{\text{MoM}}} \xrightarrow{p} Pf_{\Theta^*}$.

Given this, for every $\delta, \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n \geq N$, we have $\mathbb{P} \left(Pf_{\hat{\Theta}_n^{\text{MoM}}} - Pf_{\Theta^*} > \varepsilon \right) < \delta$ which means that for any $\eta > 0$, $\exists \varepsilon > 0$ such that $\mathbb{P} \left(\mathcal{D}(\Theta_n^{\text{MoM}}, \Theta^*) > \eta \right) < \delta$ and hence, we may conclude that $\Theta_n^{\text{MoM}} \xrightarrow{p} \Theta^*$. $\square$